

Homework & Solution

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4/27/2026

Problems 1-1

1-1. *Convert the following numbers from the given base to the other three bases listed in the table :

Solution :

Decimal	Binary	Octal	Hexadecimal
369.3125	101110001.0101	561.24	171.5
189.625	10111101.101	275.5	BD.A
214.625	11010110.101	326.5	D6.A
62407.625	1111001111000111.101	171707.5	F3C7.A

Problems 1-2

In each of the following cases, determine the radix r :

(a) $(BEE)_r = (2699)_{10}$

(b) $(365)_r = (194)_{10}$

Solution :

(a) for $(BEE)_r = (2699)_{10}$

get: $11 \times r^2 + 14 \times r^1 + 14 \times r^0 = 2699$

and $11 \times r^2 + 14 \times r - 2685 = 0$

square root : $r = 15$ or $r \approx -16.27$

Deserve : $r = 15$

(b) for $(365)_r = (194)_{10}$

get $3 \times r^2 + 6 \times r^1 + 5 \times r^0 = 194$

and $3 \times r^2 + 6 \times r - 189 = 0$

square root : $r = -9$ or $r = 7$

choose: $r = 7$

Problems 2.1

For the Boolean functions E and F, as given in the following truth table:

- (a) List the minterms and maxterms of each function.
- (b) List the minterms of $\bar{E}$ and $\bar{F}$
- (c) List the minterms of $E + F$ and $E \cdot F$.
- (d) Express E and F in sum-of-minterms algebraic form.
- (e) Simplify E and F to expressions with a minimum of literals.

XYZ	E	F
0 0 0	0	1
0 0 1	1	0
0 1 0	1	1
0 1 1	0	0
1 0 0	1	1
1 0 1	0	0
1 1 0	1	0
1 1 1	0	1

Solution :

$$a) \quad E = \sum m(1,2,4,6) = \prod M(0,3,5,7) \qquad F = \sum m(0,2,4,7) = \prod M(1,3,5,6)$$

$$c) \quad E + F = \sum m(0,1,2,4,6,7) \qquad E \cdot F = \sum m(2,4)$$

$$d) \quad E = \bar{X}\bar{Y}Z + \bar{X}Y\bar{Z} + X\bar{Y}\bar{Z} + XYZ \\ = \bar{X}\bar{Y}Z + X\bar{Z} + Y\bar{Z} \qquad F = \bar{X}\bar{Y}\bar{Z} + \bar{X}Y\bar{Z} + X\bar{Y}\bar{Z} + XYZ \\ = \bar{Y}\bar{Z} + \bar{X}\bar{Z} + XYZ$$

Problems 2-2

*Find all the prime implicants for the following Boolean functions, and determine which are essential:

(a) $F(W, X, Y, Z) = \sum m(0, 2, 5, 7, 8, 10, 12, 13, 14, 15)$

(b) $F(A, B, C, D) = \sum m(0, 2, 3, 5, 7, 8, 10, 11, 14, 15)$

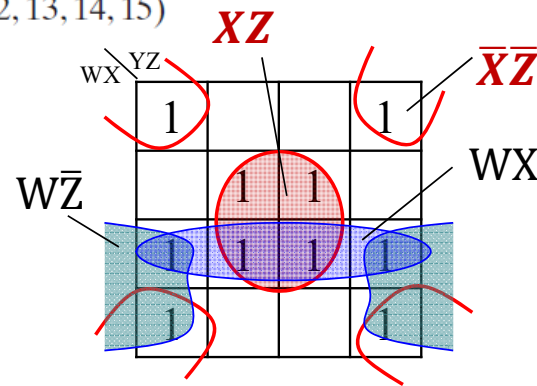
(c) $F(A, B, C, D) = \sum m(1, 3, 4, 5, 9, 10, 11, 12, 13, 14, 15)$

Solution :

a) $Prime = XZ, WX, \bar{X}\bar{Z}, W\bar{Z}$
 $Essential = XZ, \bar{X}\bar{Z}$

b) $Prime = CD, AC, \bar{B}\bar{D}, \bar{A}BD, \bar{B}C$
 $Essential = AC, \bar{B}\bar{D}, \bar{A}BD$

c) $Prime = AB, AC, AD, B\bar{C}, \bar{B}D, \bar{C}D$
 $Essential = AC, B\bar{C}, \bar{B}D$



Problems 2.3

2-3. +Prove the identity of each of the following Boolean equations, using algebraic manipulation:

(a) $ABC\bar{C} + B\bar{C}\bar{D} + BC + \bar{C}D = B + \bar{C}D$

(b) $WY + \bar{W}Y\bar{Z} + WXZ + \bar{W}X\bar{Y} = WY + \bar{W}X\bar{Z} + \bar{X}Y\bar{Z} + X\bar{Y}Z$

(c) $A\bar{D} + \bar{A}B + \bar{C}D + \bar{B}C = (\bar{A} + \bar{B} + \bar{C} + \bar{D})(A + B + C + D)$

Solution 2.3:

(a)

$$\begin{aligned} & ABC\bar{C} + B\bar{C}\bar{D} + BC + \bar{C}D \\ &= ABC\bar{C} + B(\bar{C} + \bar{D}) + BC + \bar{C}D \\ &= ABC\bar{C} + B\bar{C} + B\bar{D} + BC + \bar{C}D \\ &= ABC\bar{C} + B(\bar{C} + C) + B\bar{D} + \bar{C}D \\ &= ABC\bar{C} + B + B\bar{D} + \bar{C}D \\ &= B(1 + \bar{A}\bar{C} + \bar{D}) + \bar{C}D \\ &= B + \bar{C}D \end{aligned}$$

(c)

$$\begin{aligned} & A\bar{D} + \bar{A}B + \bar{C}D + \bar{B}C \\ &= \overline{\overline{A\bar{D} + \bar{A}B + \bar{C}D + \bar{B}C}} \\ &= \overline{(\bar{A} + D)(C + \bar{D})(A + \bar{B})\bar{B}C} \\ &= \overline{(\bar{A}C + \bar{A}\bar{D} + CD)(B + \bar{C})(A + \bar{B})} \\ &= \overline{(\bar{A}BC + \bar{A}B\bar{D} + BCD + \bar{A}\bar{C}\bar{D})(A + \bar{B})} \\ &= \overline{ABCD + \bar{A}B\bar{C}\bar{D}} \\ &= (\bar{A} + \bar{B} + \bar{C} + \bar{D})(A + B + C + D) \end{aligned}$$

Problems 2-4

*Optimize the following Boolean functions F together with the don't-care conditions d. Find all prime implicants and essential prime implicants, and apply the selection rule.

(a) $F(A, B, C) = \sum m(3, 5, 6)$, $d(A, B, C) = \sum m(0, 7)$

(b) $F(W, X, Y, Z) = \sum m(0, 2, 4, 5, 8, 14, 15)$, $d(W, X, Y, Z) = \sum m(7, 10, 13)$

(c) $F(A, B, C, D) = \sum m(4, 6, 7, 8, 12, 15)$, $d(A, B, C, D) = \sum m(2, 3, 5, 10, 11, 14)$

Solution 2.25:

(a) *Primes:* $AB, AC, BC, \bar{A}\bar{B}\bar{C}$;

Essential: AB, AC, BC ;

$$F = AB + AC + BC$$

(b) *Primes:* $\bar{X}\bar{Z}, XZ, \bar{W}X\bar{Y}, WXY, \bar{W}\bar{Y}\bar{Z}, WY\bar{Z}$

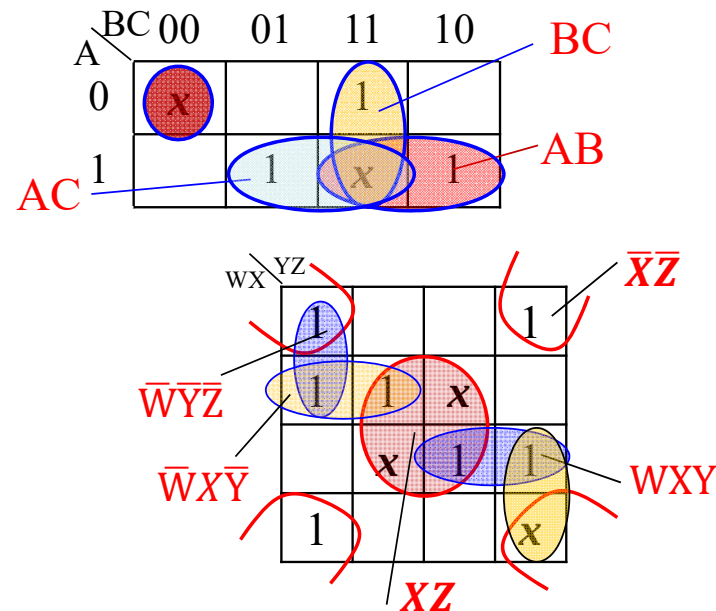
Essential: $\bar{X}\bar{Z}$

$$F = \bar{X}\bar{Z} + \bar{W}X\bar{Y} + WXY$$

(c) *Primes:* $\bar{A}B, C, A\bar{D}, B\bar{D}$

Essential: $C, A\bar{D}$

$$F = C + A\bar{D} + (B\bar{D} \text{ or } \bar{A}B)$$



Problems 3-1

3-1. A hierarchical component with the function is to be used along with inverters to implement the following equation:

$$H = \bar{X}Y + XZ$$

$$G = \bar{A}\bar{B}C + \bar{A}BD + A\bar{B}\bar{C} + AB\bar{D}$$

The overall circuit can be obtained by using **Shannon's expansion** theorem,

$$F = \bar{X} \cdot F_0(X) + X \cdot F_1(X)$$

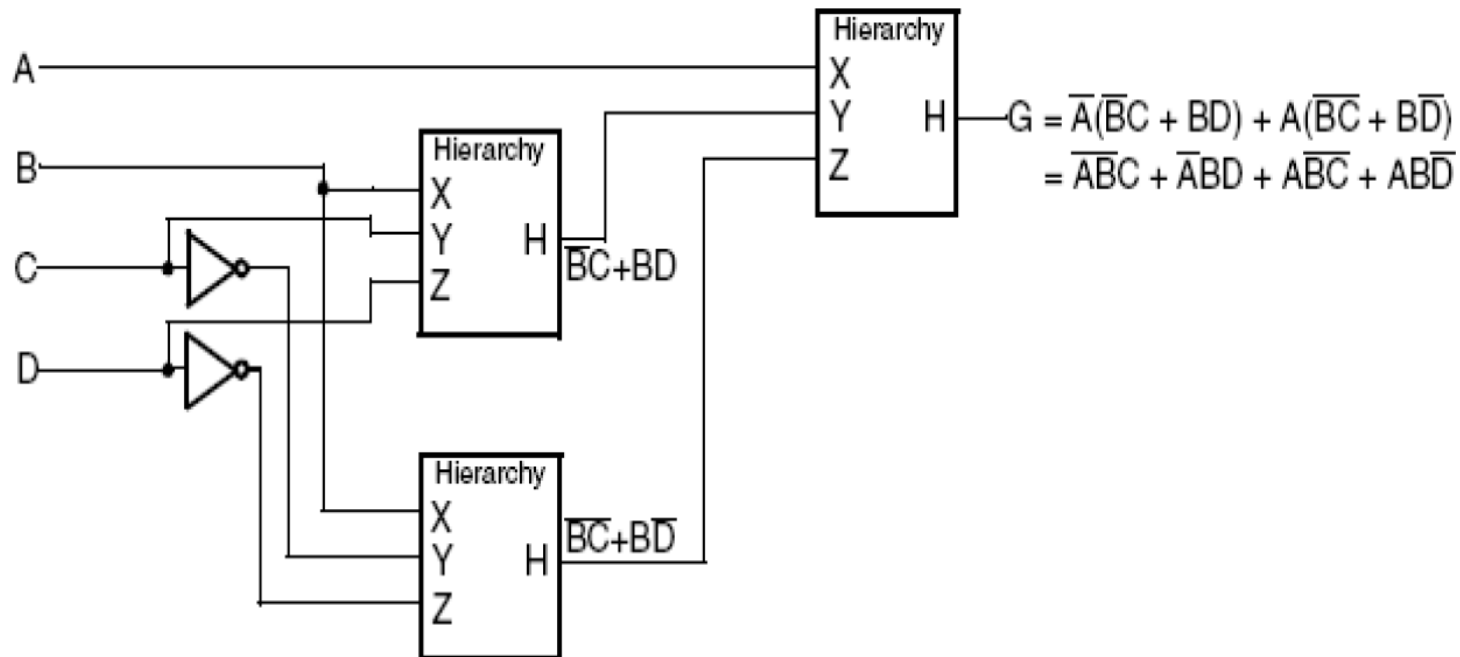
where $F_0(X)$ is F evaluated with variable $X = 0$ and $F_1(X)$ is F evaluated with variable $X = 1$. This expansion F can be implemented with function H by letting $Y = F_0$ and $Z = F_1$. The expansion theorem can then be applied to each of F_0 and F_1 using a variable in each, preferably one that appears in both true and complemented form. The process can then be repeated until all F_i 's are single literals or constants.

For G , use $X = A$ to find G_0 and G_1 and then use $X = B$ for G_0 and G_1 . Draw the top-level diagram for G using H as a hierarchical component.

Solution 3-1:

$$\begin{aligned} G &= \bar{A}\bar{B}C + \bar{A}BD + A\bar{B}\bar{C} + AB\bar{D} \\ &= \bar{A}(\bar{B}C + BD) + A(\bar{B}\bar{C} + B\bar{D}) \end{aligned}$$

$$H = \bar{X}Y + XZ$$



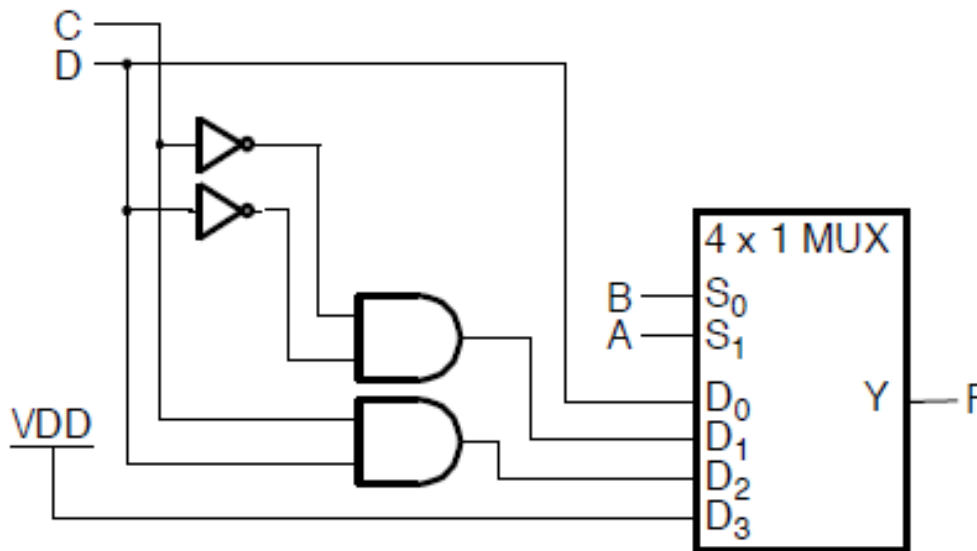
Problems 3-2

3-2. *Implement the Boolean function

$$F(A, B, C, D) = \sum m(1, 3, 4, 11, 12, 13, 14, 15)$$

with a 4-to-1-line multiplexer and external gates. Connect inputs A and B to the selection lines. The input requirements for the four data lines will be a function of the variables C and D. The values of these variables are obtained by expressing F as a function of C and D for each of the four cases when AB = 00, 01, 10, and 11. These functions must be implemented with external gates.

Solution 3.44:

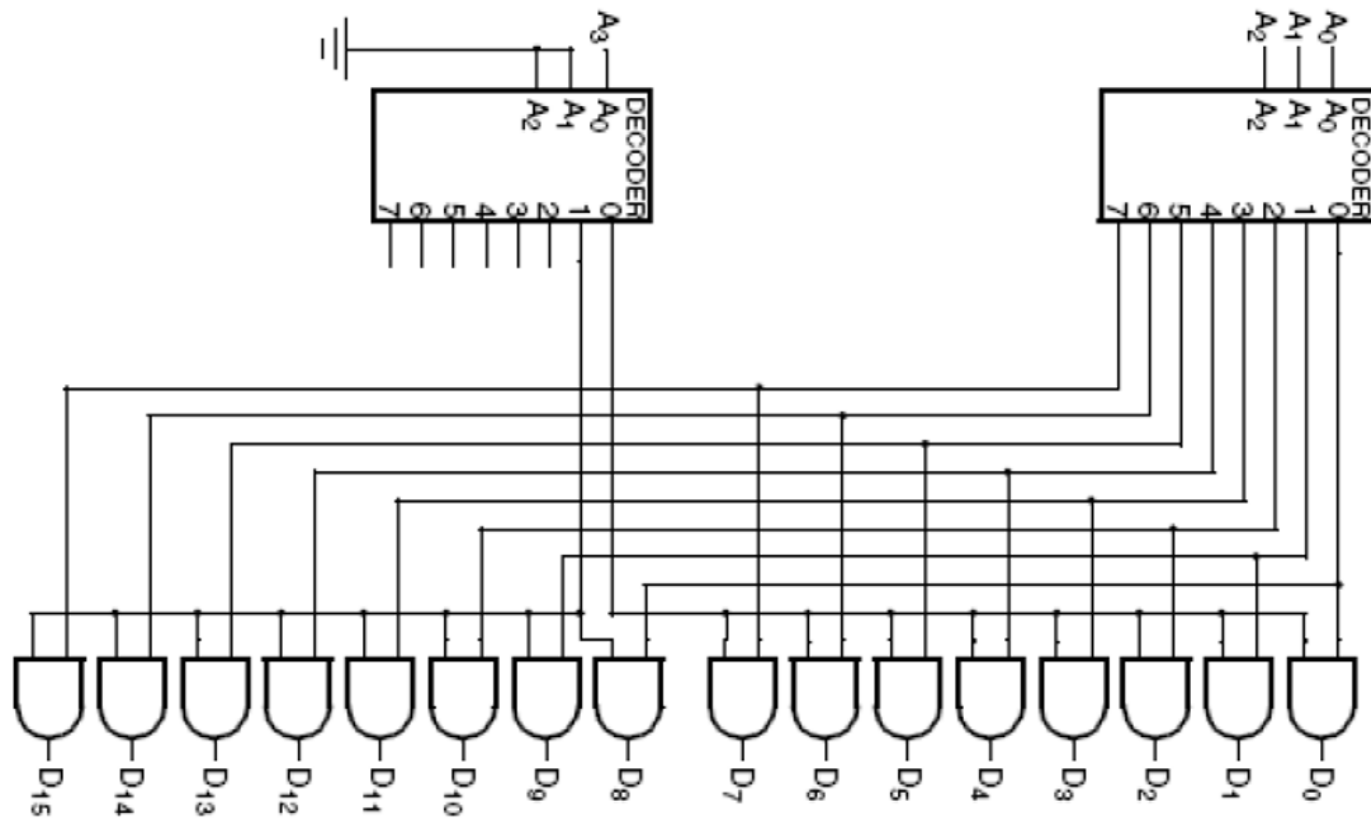


A	B	C	D	F	
0	0	0	0	0	F = D
0	0	0	1	1	
0	0	1	0	0	
0	0	1	1	1	
0	1	0	0	1	F = $\overline{C}D$
0	1	0	1	0	
0	1	1	0	0	
0	1	1	1	0	
1	0	0	0	0	F = CD
1	0	0	1	0	
1	0	1	0	0	
1	0	1	1	1	
1	1	0	0	1	F = 1
1	1	0	1	1	
1	1	1	0	1	
1	1	1	1	1	

Problems 3-3

3-3. (a) Design a 4-to-16-line decoder using two 3-to-8-line decoders and 16 2-input AND gates.

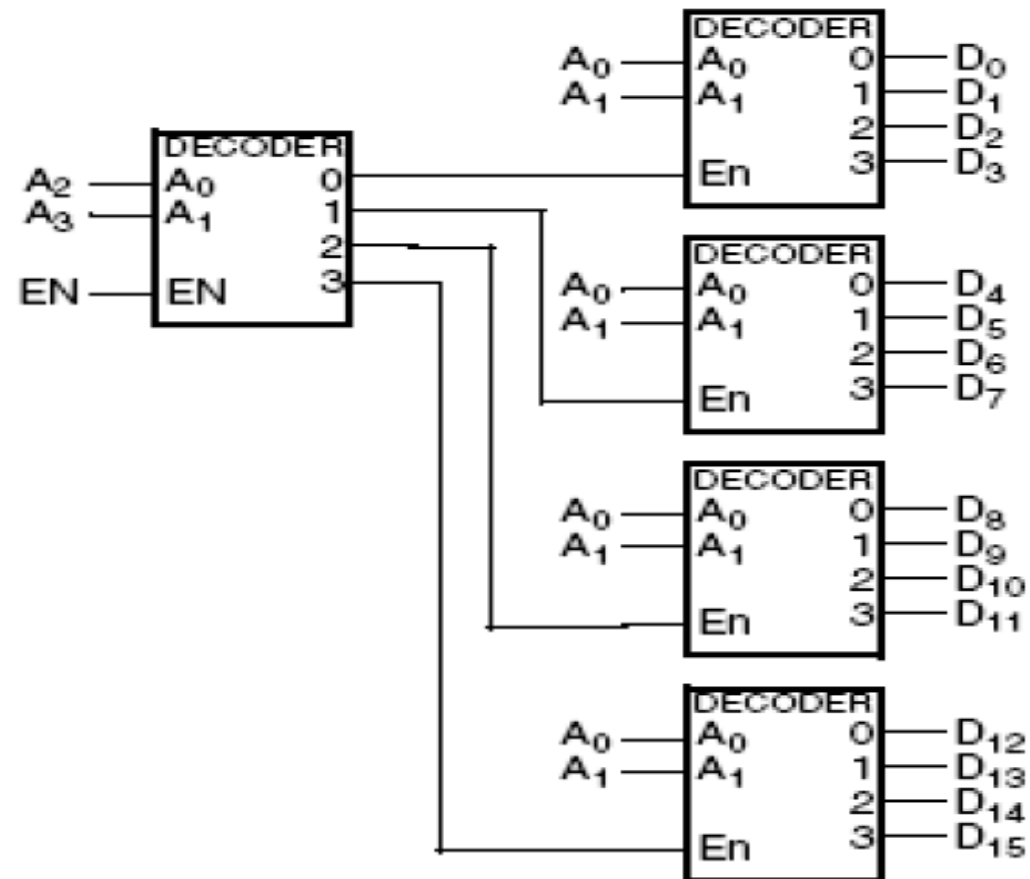
Solution :



Problems 3-3

(b) Design a 4-to-16-line decoder with enable using five 2-to-4-line decoders with enable as shown in Figure 3-16.

Solution :



Problems 3-4

3-4. Design a combinational circuit that compares two 4-bit unsigned numbers A and B to see whether B is greater than A. The circuit has one output X, so that $X = 1$ if $A < B$ and $X = 0$ if $A \geq B$.

Solution :

a. Design 1 bit comparator:

$$\begin{aligned} A_i > B_i & \quad Z_i = A_i \overline{B_i} \\ A_i = B_i & \quad Y_i = A_i \overline{B_i} + B_i \overline{A_i} \\ A_i < B_i & \quad W_i = B_i \overline{A_i} \end{aligned}$$

输入		Zi	Yi	Wi
Ai	Bi	Ai > Bi	Ai = Bi	Ai < Bi
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

b. 4-bit comparator compares from most significant to lower with bit-by-bit processing:

if $A_i < B_i$, $(\overline{A_i} B_i = 1)$ and all $j > i, A_j = B_j$ $(\overline{A_j} \overline{B_j} + A_j B_j = 1)$, then $A < B$.

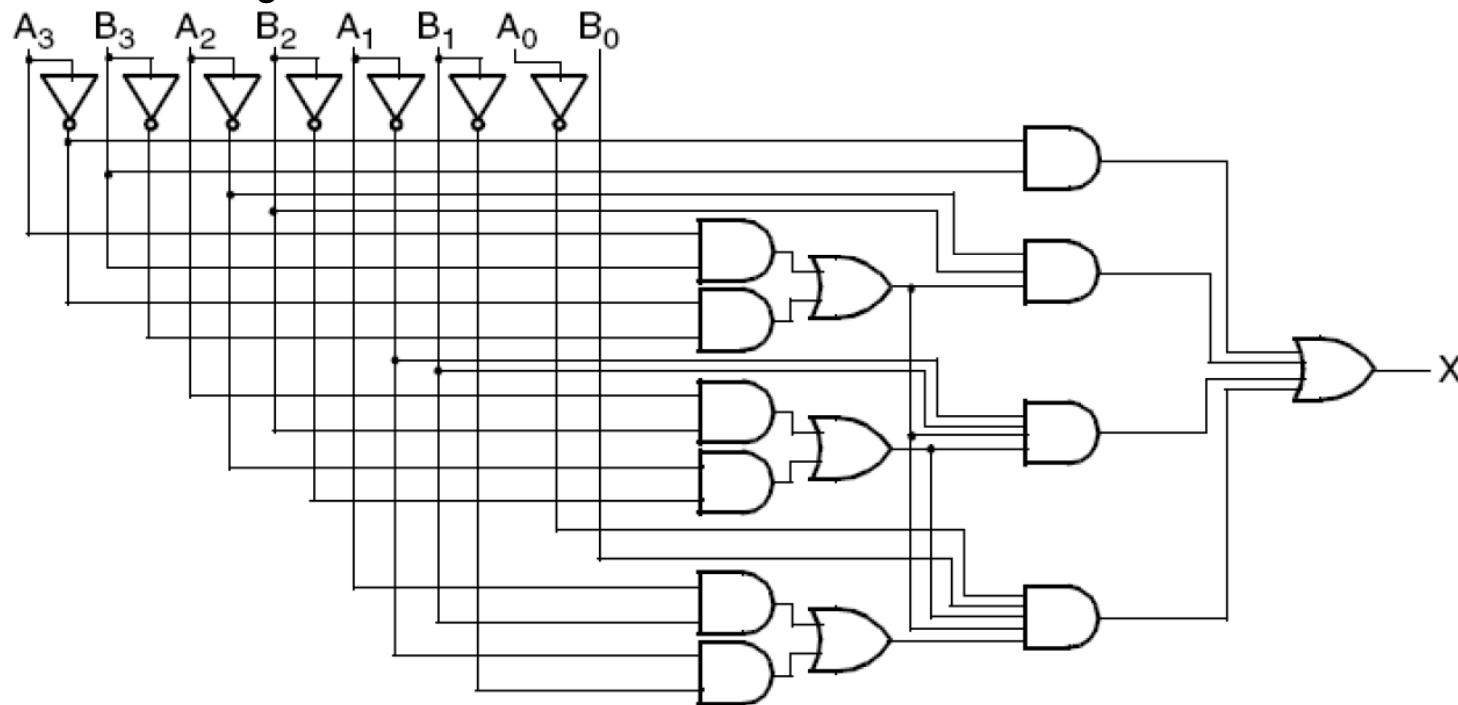
$A_3 < B_3$	$A_3 = B_3$ & $A_2 < B_2$	$A_3 = B_3, A_2 = B_2$ & $A_1 < B_1$	$A_3 = B_3, A_2 = B_2, A_1 = 1$ & $A_0 < B_0$	X
1	x	x	x	1
	1	x	x	1
		1	x	1
			1	1
				0

Solution 3-4

c. expression

$$X = \bar{A}_3 B_3 + (A_3 B_3 + \bar{A}_3 \bar{B}_3) \bar{A}_2 B_2 + (A_3 B_3 + \bar{A}_3 \bar{B}_3) (A_2 B_2 + \bar{A}_2 \bar{B}_2) \bar{A}_1 B_1 \\ + (A_3 B_3 + \bar{A}_3 \bar{B}_3) (A_2 B_2 + \bar{A}_2 \bar{B}_2) (A_1 B_1 + \bar{A}_1 \bar{B}_1) \bar{A}_0 B_0$$

d. Circuit diagram



- 解法二：用串行加法器构造电路A-B，即在B上加非门， C_0 接1，在 C_4 上加非门作为X。
-
- 解法三：用串行加法器构造电路B-A-1，即在A上加非门， C_0 接0，将 C_4 作为X。

Problem 4-1

3.20 [5] <\$3.5> What decimal number does the bit pattern $0 \times 0C000000$ represent if it is a two's complement integer? An unsigned integer?

3.21 [10] <\$3.5> If the bit pattern $0 \times 0C000000$ is placed into the Instruction Register, what MIPS instruction will be executed?

3.22 [10] <\$3.5> What decimal number does the bit pattern $0 \times 0C000000$ represent if it is a floating point number? Use the IEEE 754 standard.

3.23 [10] <\$3.5> Write down the binary representation of the decimal number 63.25 assuming the IEEE 754 single precision format.

3.24 [10] <\$3.5> Write down the binary representation of the decimal number 63.25 assuming the IEEE 754 double precision format.

Solution:

- 3.20 $12 \times 16^6 = 201326592$ in both cases.
- 3.21 jal $0x00000000$
- 3.22 $0x0C000000 = 0000\ 1100\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000$
sign is positive, $\text{exp} = 24 - 127 = -103$, fraction = 0, so answer = 1.0×2^{-103}
- 3.23 $63.25 = 111111.01 = 1.1111101 \times 2^5$
sign = positive, $\text{exp} = 127 + 5 = 132$
so IEEE 754 single format = $0\ 1000\ 0100\ 1111\ 1010\ 0000\ 0000\ 0000\ 0000\ 0000 = 0x427D0000$
- 3.24 sign = positive, $\text{exp} = 1023 + 5 = 1028$
so IEEE 754 double format = $0\ 100\ 0000\ 0100\ 1111\ 1010\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000\ 0000$
 $= 0x404FA00000000000$

Problem 4-2

3.27 [20] <§3.5> IEEE 754-2008 contains a half precision that is only 16 bits wide. The leftmost bit is still the sign bit, the exponent is 5 bits wide and has a bias of 15, and the mantissa is 10 bits long. A hidden 1 is assumed. Write down the bit pattern to represent -1.5625×10^{-1} assuming a version of this format, which uses an excess-16 format to store the exponent. Comment on how the range and accuracy of this 16-bit floating point format compares to the single precision IEEE 754 standard.

Solution:

$$-1.5625 \times 10^{-1} = -0.00101 = -1.01 \times 2^{-3}$$

$$\text{exponent} = -3 + 15 = 12 : 01100,$$

$$\text{fraction} = -0.0100000000$$

$$\text{answer: } 1\ 01100\ 0100000000$$

Problems 5-1

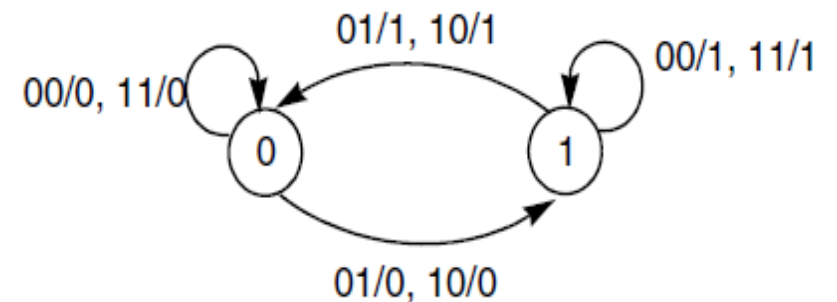
5-1. A sequential circuit has one flip-flop Q , two inputs X and Y , and one output S . The circuit consists of a D flip-flop with S as its output and logic implementing the function

$$D = X \oplus Y \oplus S$$

with D as the input to the D flip-flop. Derive the state table and state diagram of the sequential circuit.

Solution :

Present state	Inputs		Next state	Output
Q	X	Y	Q	S
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	0
1	0	0	1	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



Format: XY/S

Problems 5-2

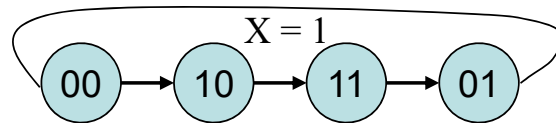
*Design a sequential circuit with two D flip-flops A and B and one input X. When $X = 0$, the state of the circuit remains the same. When $X = 1$, the circuit goes through the state transitions from 00 to 10 to 11 to 01, back to 00, and then repeats.

Solution:

(a) State diagrams :

(b) State table :

Present state		Input	Next state	
A	B		A	B
0	0	0	0	0
0	0	1	1	0
0	1	0	0	1
0	1	1	0	0
1	0	0	1	0
1	0	1	1	1
1	1	0	1	1
1	1	1	0	1



(c) State/Input equations:

D_A		B	
		1	
A	1	1	1
		X	

$$D_A = A\bar{X} + \bar{B}X$$

D_B		B	
			1
A		1	1
		X	

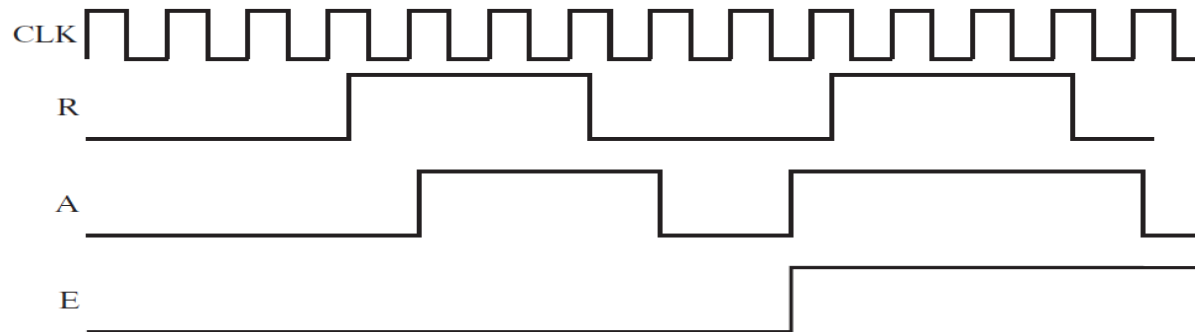
$$D_B = AX + B\bar{X}$$

(d) Logic diagram:

Skip, Draw your own.

Problems 5-3

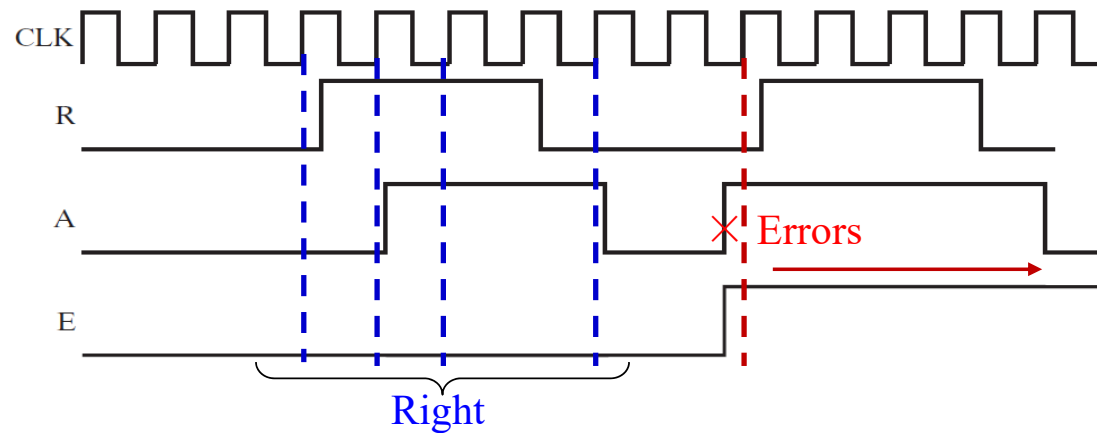
A pair of signals Request (R) and Acknowledge (A) is used to coordinate transactions between a CPU and its I/O system. The interaction of these signals is often referred to as a “handshake.” These signals are synchronous with the clock and, for a transaction, are to have their transitions always appear in the order shown in Figure 4-53. A handshake checker is to be designed that will verify the transition order. The checker has inputs, R and A , asynchronous reset signal, RESET, and output, Error (E). If the transitions in a handshake are in order, $E = 0$. If the transitions are out of order, then E becomes 1 and remains at 1 until the asynchronous reset signal (RESET = 1) is applied to the CPU.



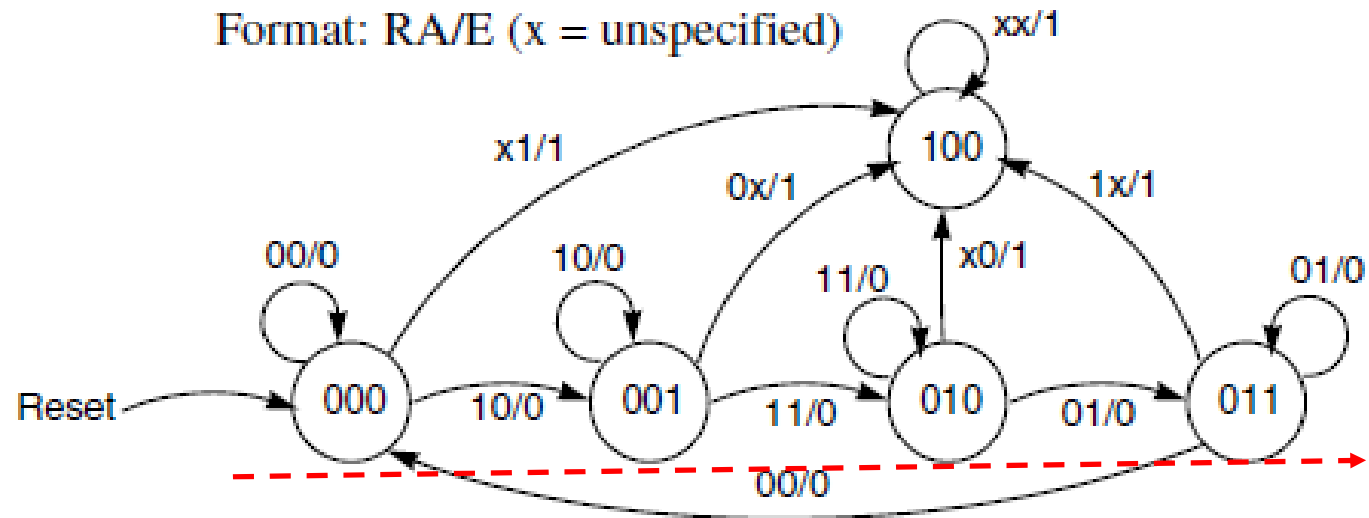
- (a) Find the state diagram for the handshake checker.
- (b) Find the state table for the handshake checker.

Solution 5-3

- (a) State diagram
= 00 10 11 01 is right



Format: RA/E (x = unspecified)



Solution Continue before:

(b) State table

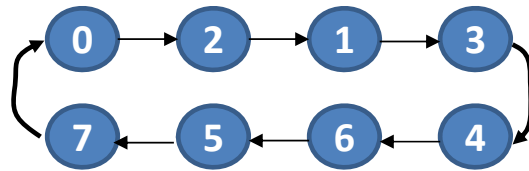
Present state			Inputs		Next state			Output
<i>B</i>	<i>C</i>	<i>D</i>	<i>R</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
0	0	0	0	0	0	0	0	0
0	0	0	0	1	1	0	0	1
0	0	0	1	0	0	0	1	0
0	0	0	1	1	1	0	0	1
0	0	1	0	0	1	0	0	1
0	0	1	0	1	1	0	0	1
0	0	1	1	0	0	0	1	0
0	0	1	1	1	0	1	0	0
0	1	0	0	0	1	0	0	1
0	1	0	0	1	0	1	1	0
0	1	0	1	0	1	0	0	1
0	1	0	1	1	0	1	0	0
0	1	1	0	0	0	0	0	0
0	1	1	0	1	0	1	1	0
0	1	1	1	0	1	0	0	1
0	1	1	1	1	1	0	0	1
1	0	0	0	0	1	0	0	1
1	0	0	0	1	1	0	0	1
1	0	0	1	0	1	0	0	1
1	0	0	1	1	1	0	0	1

Problems 5-4

Use *D*-type flip-flops and gates to design a counter with the following repeated binary sequence: 0, 2, 1, 3, 4, 6, 5, 7.

Solution :

(a) State diagram:



(b) State Table:

Present state	Nest state
C B A	C B A
0 0 0	0 1 0
0 0 1	0 1 1
0 1 0	0 0 1
0 1 1	1 0 0
1 0 0	1 1 0
1 0 1	1 1 1
1 1 0	1 0 1
1 1 1	0 0 0

(c) State equation and input equation

$$D_C = C_{t+1} = \overline{B}C + \overline{A}C + AB\overline{C}$$

$$D_B = B_{t+1} = \overline{B}$$

$$D_A = A_{t+1} = A\overline{B} + \overline{A}B = A \oplus B$$

(d) The circuit diagram

Skip, Do it yourself!

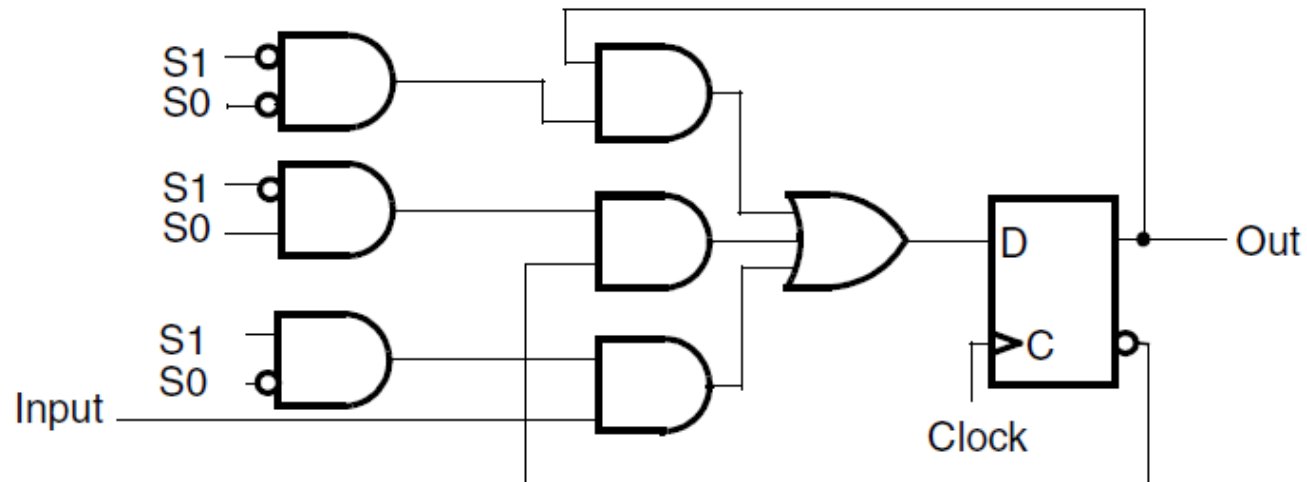
Problems 5-5

Draw the logic diagram of a 4-bit register with mode selection inputs S_1 and S_0 . The register is to be operated according to the function table below.

S_1	S_0	Register Operation
0	0	No change
0	1	Complement output
1	0	Load parallel data
1	1	Clear register to 0

Solution:

“与”门使能直接构成多路选择器，选择控制是 S_1S_0 ： The basic cell of the register is as follows



Problems 5-6

A register cell is to be designed for an 8-bit register A that has the following register transfer functions:

$$C_0: A \leftarrow A \cap B, C_1: A \leftarrow A \cup \bar{B}$$

Find optimum logic using AND, OR, and NOT gates for the D input to the D flip-flop in the cell.

Solution:

典型寄存器传输操作控制:

寄存器传输 + 微操作

$C_0 = 1$ 时, 与门1使能 $A \cap B$ 传输进或门;

$C_0 = 0$ 时, $A \cap B$ 传输禁止传输。

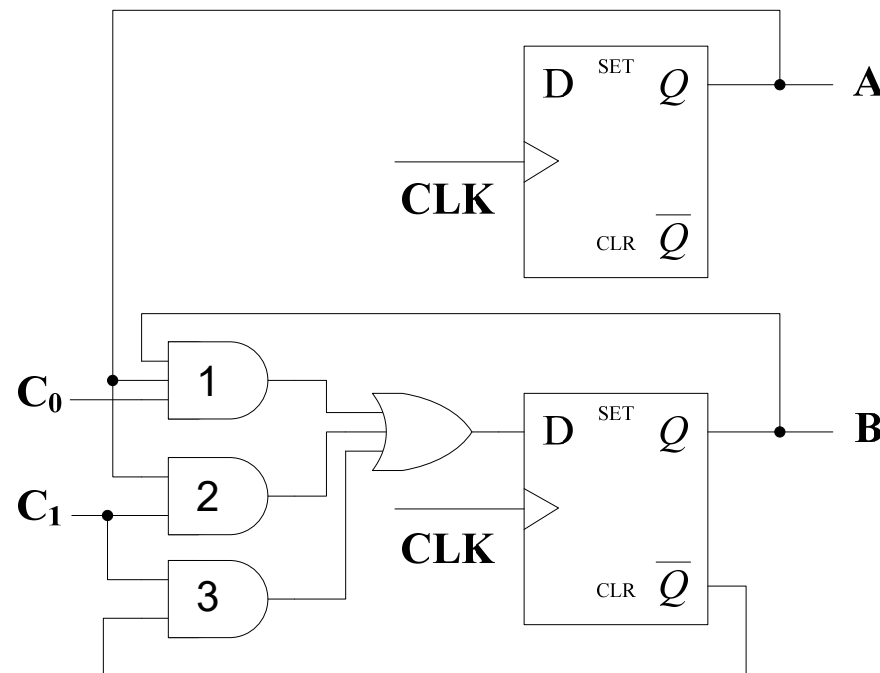
$C_1 = 1$ 时, 与门2~3使能 A 、 $\bar{B}$ 传输进入
或门运算 $A \cup \bar{B}$;

$C_1 = 0$ 时, 与门2~3禁止 A 、 $\bar{B}$ 传输。

问: 此电路设计合理吗?

提示: $C_0 C_1 = 11$, ??

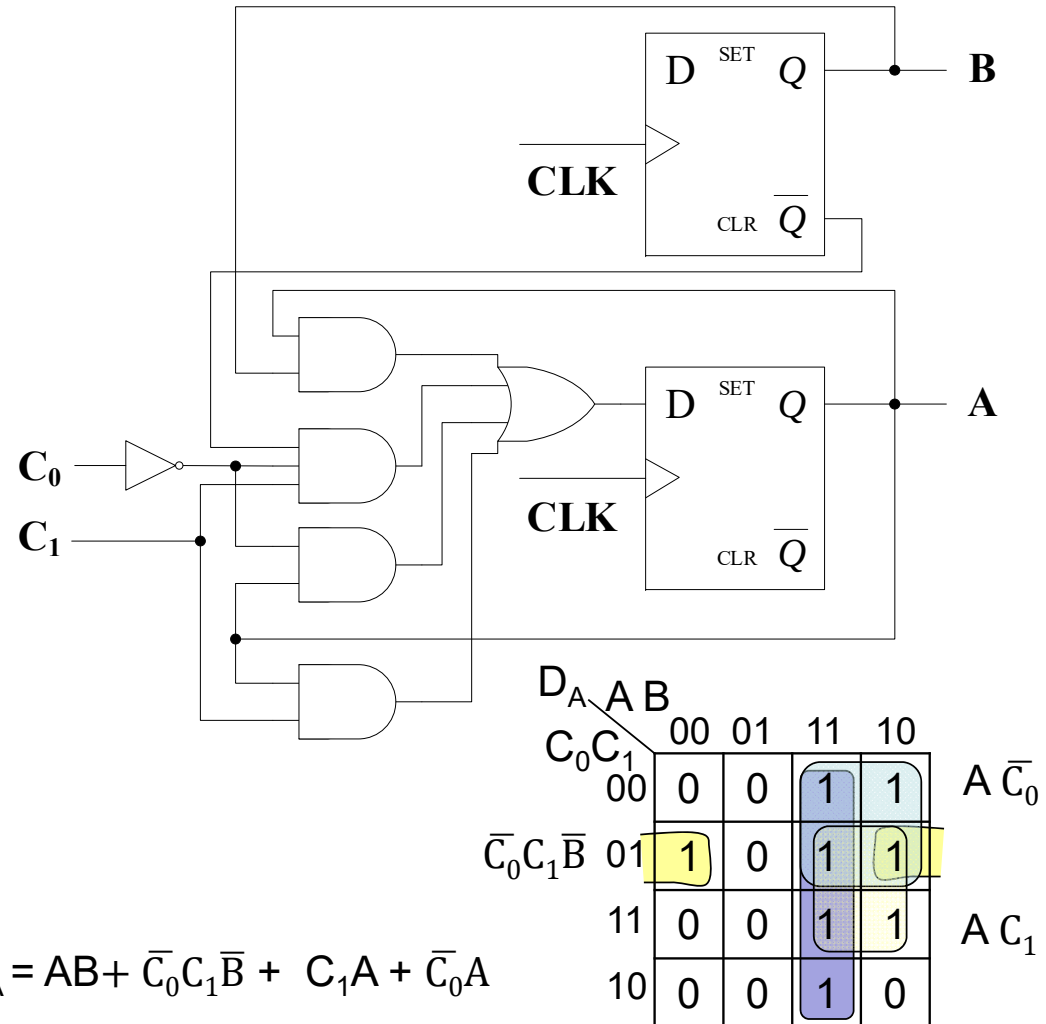
$C_0 C_1 = 00$, ??



改进

$C_0: A \leftarrow A \cap B, C_1: A \leftarrow A \cup \bar{B}$

C_0	C_1	A	B	$A_{(t+1)}$
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1



Problems 5-7

*Two register transfer statements are given (otherwise, R1 is unchanged):

C1: $R1 \leftarrow R1 + R2$ Add R2 to R1

$\overline{C1}$ C2: $R1 \leftarrow R1 + 1$ Increment R1

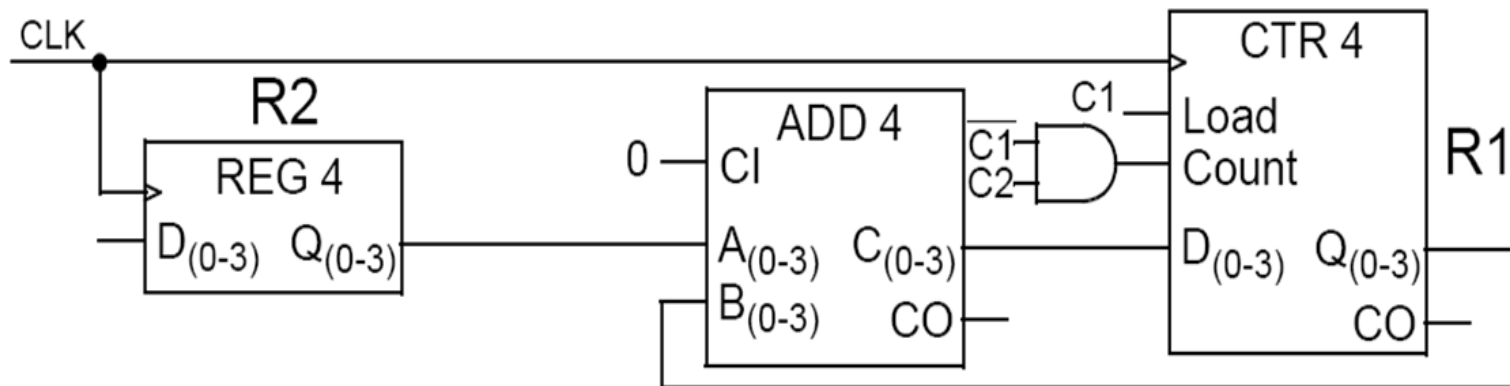
(a) Using a 4-bit counter with parallel load and a 4-bit adder, draw the logic diagram that implements these register transfers.

(b) Repeat part (a) using a 4-bit adder as in Figure 3-43 plus external gates as needed. Compare with the implementation in part (a).

Solution :

(a) $\overline{C1}$ C2=11 时, Load=C1=0, 禁止加载, CTR4计数, 实现Increment R1;

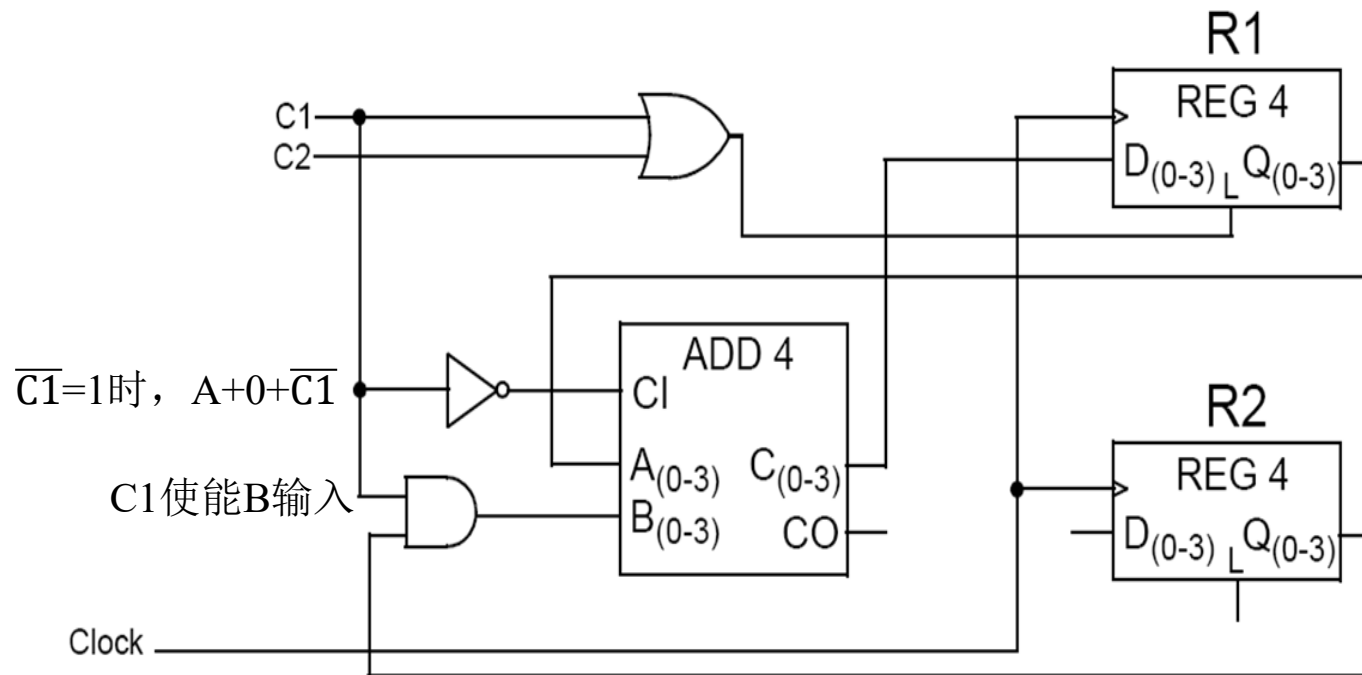
C1C2 = 1x时, CTR4计数禁止, Load=C1=1, 加载ADD4输出, 实现Add R2 to R1。



Solution Continue before:

(b) Repeat part (a) using a 4-bit adder as in Figure 3-43 plus external gates as needed. Compare with the implementation in part (a).

$\overline{C1}C2=11$ 时, ADD4输入: $CI=\overline{C1}=1$, B被与门禁止输入, 此时实现 $R1 \leftarrow R1+1$;
 $C1C2=1x$ 时, ADD4输入: $CI=\overline{C1}=0$, 与门使能B正常输入, 实现 $R1 \leftarrow R1+R2$ 。



Problems 5-8

*A sequential circuit is shown in Figure 4-49. The timing parameters for the gates and flip-flops are as follows:

Inverter: $t_{pd} = 0.01$ ns; XOR gate: $t_{pd} = 0.04$ ns

Flip-flop: $t_{pd} = 0.08$ ns, $t_s = 0.02$ ns, and $t_h = 0.01$ ns

(a) Find the longest path delay from an external circuit input passing through gates only to an external circuit output.

(b) Find the longest path delay in the circuit from an external input to positive clock edge.

(c) Find the longest path delay from positive clock edge to output.

(d) Find the longest path delay from positive clock edge to positive clock edge.

(e) Determine the maximum frequency of operation of the circuit in megahertz (MHz).

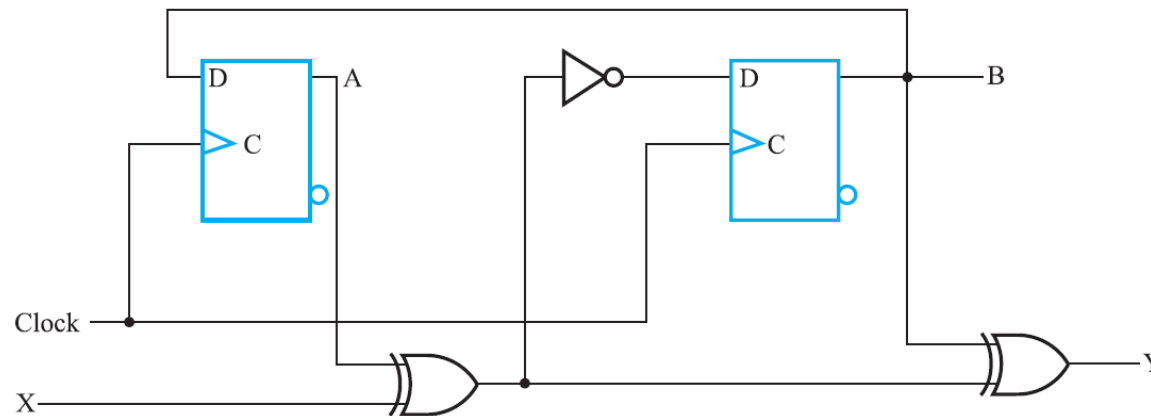
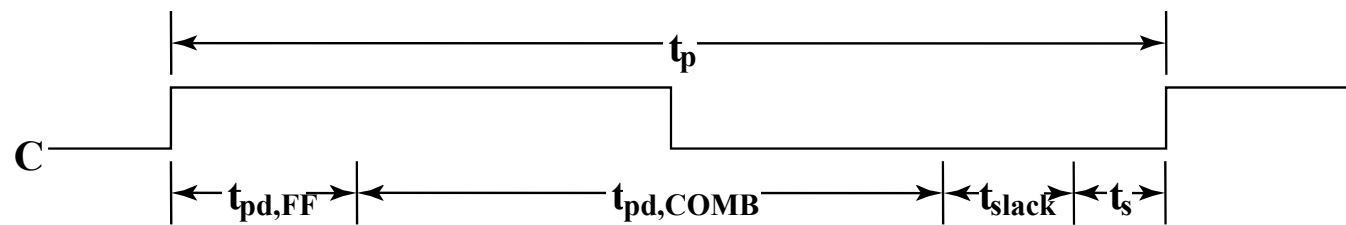
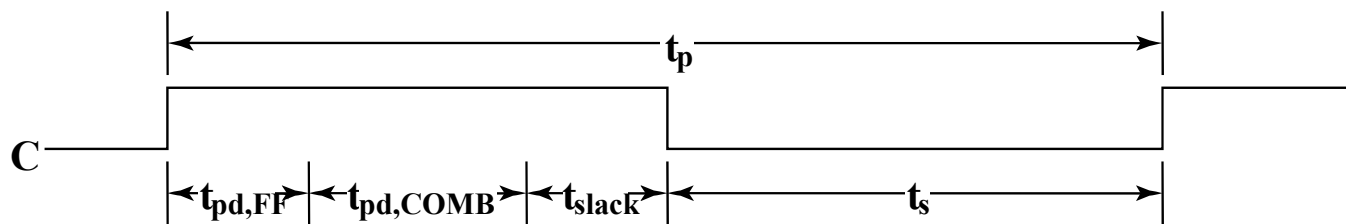


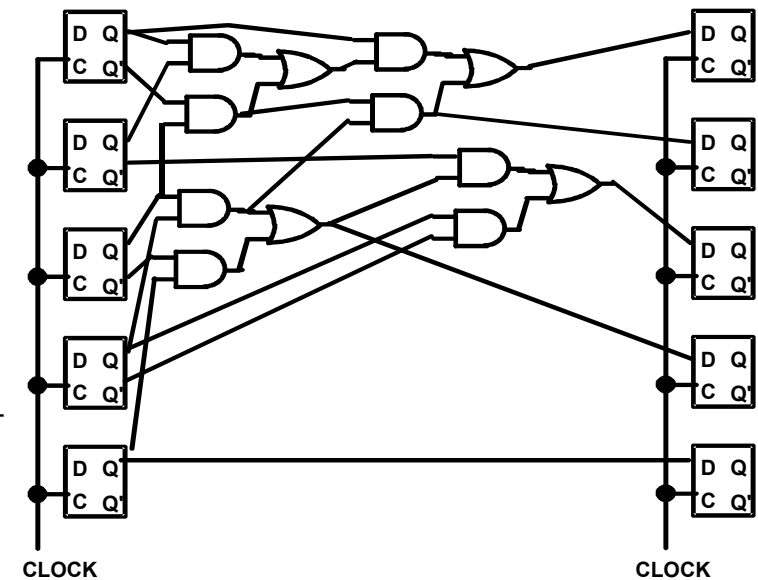
Figure 4-49



(a) Edge-triggered (positive edge)

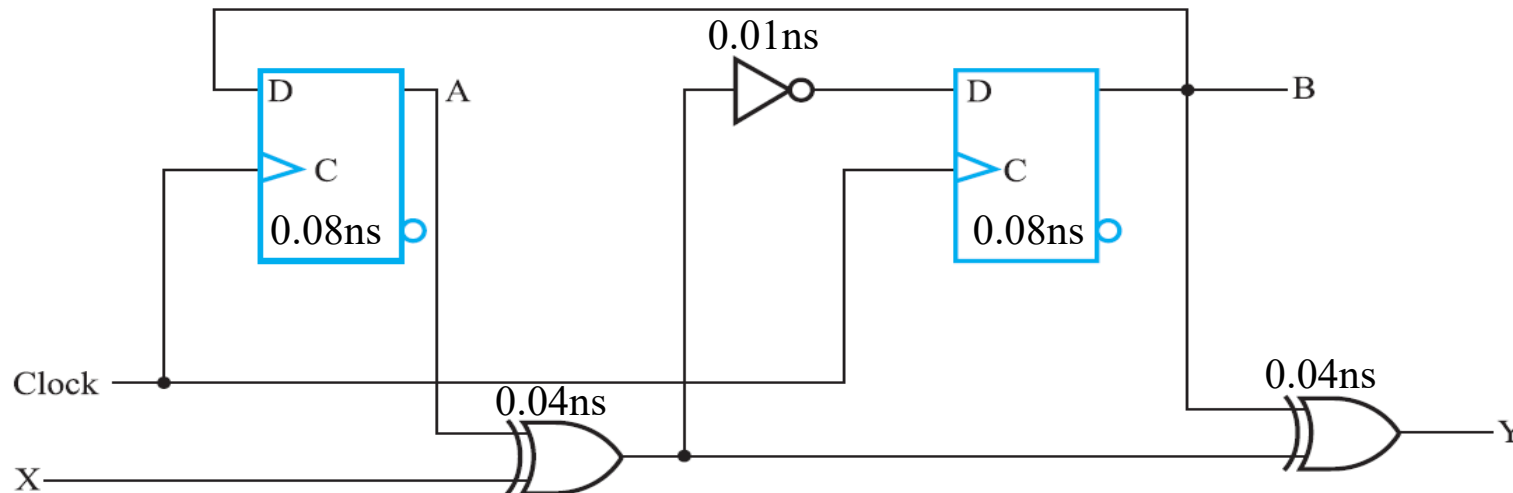


(b) Pulse-triggered (negative pulse)



Solution

$$\begin{aligned} t_{pd} &= 0.08 \\ t_s &= 0.02 \\ t_h &= 0.01 \end{aligned}$$



- (a) $t_{\text{delay}} = t_{\text{pdXOR}} + t_{\text{pdXOR}} = 0.04 + 0.04 = 0.08 \text{ ns}$
- (b) $t_{\text{delay}} = t_{\text{pdXOR}} + t_{\text{pdINV}} + t_{\text{sFF}} = 0.04 + 0.01 + 0.02 = 0.07 \text{ ns}$
- (c) $t_{\text{delay}} = t_{\text{pdFF}} + 2 t_{\text{pdXOR}} = 0.08 + 2(0.04) = 0.16 \text{ ns}$
- (d) $t_{\text{delay-clock edge to clock edge}} = t_{\text{pdFF}} + t_{\text{pdXOR}} + t_{\text{pdINV}} + t_{\text{sFF}}$
 $= 0.08 + 0.04 + 0.01 + 0.02 = 0.15 \text{ ns}$
- (e) $f_{\text{max}} = 1/t_{\text{delay-clock edge to clock edge}} = 1/0.15 \text{ ns} = 6.67 \text{ GHz}$

Comment: The clock frequency may need to be lower due to other delay paths that pass outside of the circuit into its environment. Calculation of this frequency cannot be performed in this case since data for paths through the environment is not provided.